

1 Solution — GIAM 6.3

Problem 6

1.1 Problem

The relation Q from Problem 5 — defined on $\mathbb{Z} \times \mathbb{Z}^*$ by $(a, b) Q (c, d) \iff ad = bc$ — partitions $\mathbb{Z} \times \mathbb{Z}^*$ into equivalence classes. Explain why this is an “interesting” set.

1.2 Brief Answer

The equivalence classes *are the rational numbers*. There is a natural bijection

$$(\mathbb{Z} \times \mathbb{Z}^*)/Q \longleftrightarrow \mathbb{Q}, \quad [(a, b)]_Q \longleftrightarrow \frac{a}{b},$$

so the quotient *is* $\mathbb{Q}$ — the rational numbers, constructed entirely from $\mathbb{Z}$ and the equivalence relation Q .

1.3 What an Equivalence Class Looks Like

For any $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$, the class is

$$[(a, b)]_Q = \{ (c, d) \in \mathbb{Z} \times \mathbb{Z}^* : ad = bc \}.$$

The condition $ad = bc$ says exactly “ $a/b = c/d$ ” — i.e., (c, d) is some non-zero scalar multiple of (a, b) . If $\gcd(|a|, |b|) = 1$ (so a/b is in lowest terms), the class is

$$[(a, b)]_Q = \{ (ka, kb) : k \in \mathbb{Z}^* \}.$$

For example,

$$[(1, 2)]_Q = \{ \dots, (-2, -4), (-1, -2), (1, 2), (2, 4), (3, 6), \dots \}$$

— every (positive or negative) scaling of $(1, 2)$, all of them representing the same rational $\frac{1}{2}$.

1.4 Identifying Classes with $\mathbb{Q}$

Define

$$\varphi : (\mathbb{Z} \times \mathbb{Z}^*)/Q \longrightarrow \mathbb{Q}, \quad \varphi([(a, b)]_Q) = \frac{a}{b}.$$

This “interpret a pair as a fraction” map is a bijection:

- **Well-defined.** If $(a, b) Q (c, d)$, then $ad = bc$, hence $a/b = c/d$ in $\mathbb{Q}$ (divide both sides by $bd \neq 0$). So every representative of the class lands on the same rational. ✓
- **Injective.** If $a/b = c/d$ in $\mathbb{Q}$, then cross-multiplying gives $ad = bc$, i.e. $(a, b) Q (c, d)$, so $[(a, b)]_Q = [(c, d)]_Q$. ✓
- **Surjective.** Any $q \in \mathbb{Q}$ can be written as a/b with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}^*$; that pair lies in $\varphi^{-1}(q)$. ✓

So the quotient set is in one-to-one correspondence with $\mathbb{Q}$.

1.5 A Few Concrete Classes

Rational	A representative	The whole class
0	$(0, 1)$	$\{(0, d) : d \in \mathbb{Z}^*\}$
1	$(1, 1)$	$\{(k, k) : k \in \mathbb{Z}^*\}$
$\frac{1}{2}$	$(1, 2)$	$\{(k, 2k) : k \in \mathbb{Z}^*\}$
$-\frac{2}{3}$	$(-2, 3)$	$\{(-2k, 3k) : k \in \mathbb{Z}^*\}$

Table 1.1

Each rational has infinitely many representative pairs — one per non-zero scaling — and they all collapse into a single equivalence class.

1.6 Remark — More Than a Bijection of Sets

The identification with $\mathbb{Q}$ is much more than a set bijection: the quotient inherits all of the arithmetic of $\mathbb{Q}$. Define

- **Addition:** $[(a, b)]_Q + [(c, d)]_Q := [(ad + bc, bd)]_Q$,

- **Multiplication:** $[(a, b)]_Q \cdot [(c, d)]_Q := [(ac, bd)]_Q.$

These are exactly the schoolbook rules

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd},$$

now read at the level of equivalence classes. One checks that both operations are well-defined (the result does not depend on the choice of representative), and the quotient becomes a *field* — the field $\mathbb{Q}$.

So Problems 5 and 6 together carry out the standard construction of the rational numbers from the integers, using nothing beyond equivalence relations and quotient sets. The same recipe with $\mathbb{Z}$ replaced by any integral domain R yields the **field of fractions** $\text{Frac}(R)$ — the smallest field containing R .

That is what makes this set of equivalence classes interesting: it is the bridge from the integers to the rationals, and from any integral domain to its field of fractions.